

Exam
Data Driven Decision Making
05.08.2025 (Summer 2025)

Study program

- ☐ **Wirtschaftsinformatik (Master)**
- ☐ **Technologieorientiertes Management (Master)**
- ☐ **Wirtschaftsingenieurwesen (Master), Specialization: _____**
- ☐ **Other: _____**

Personal data

Last name: _____

First name: _____

Matr. no.: _____ **Semester:** _____

- ☐ 1. attempt ☐ 2. attempt ☐ 3. attempt

Instructions

- Please check the **completeness** of the exam (7 pages incl. cover) immediately upon receipt. Please write your matriculation number **on each page!**
- Only use the **provided solution fields** and the corresponding **back pages** or **stamped blank pages** for the solution.
- Only **document-proof writing and drawing instruments** are permitted as aids. Answers written in pencil cannot be assessed.
- Answers **without sufficient justification** will result in points being deducted.
- Answers are allowed in **German or English**.
- The working time is **60 minutes**.
- A total of **60 points** can be achieved.

Results

[illegible]

Exercise 1 (2 points)

What are the two advantages of a priori optimization as opposed to dynamic decision-making?

Exercise 2 (6 points)

Travel times are observed on a road segment and assigned to the hour in which the observation was made. The expected travel time for a given hour is calculated from the average of all observations made during that hour. What is meant by the FIFO condition regarding average travel times in consecutive hours? When is the FIFO condition violated? How can the FIFO condition be?

Exercise 3 (4 points)

In a dynamic stochastic inventory problem without prediction, how can you ensure that orders are placed at all? Which category of ADP methods would you classify your proposed solution under?

Exercise 4 (4 points)

ADP methods such as rollout or value function approximation simulate possible developments based on post-decision states. In such a simulation, the stochastic transitions are drawn from an externally given distribution. Outline the procedure for an exponentially distributed distribution.

Exercise 5 (4 points)

Explain in keywords the difference between (filter-based) feature selection and dimensionality reduction techniques. Give two specific examples for each.

Exercise 6 (8 points)

You are looking at a dynamic inventory problem. Name the external process, the decision process, the state space, and the reward.

Exercise 7 (6 points)

Specify the components of a Markov decision process.

Exercise 8 (4 points)

Explain the function and prerequisites for the applicability of the Hurwicz criterion in optimization.

Exercise 9 (10 points)

You have learned about five ADP method classes. Name these classes and state one substantial disadvantage of using each class.

Exercise 10 (12 points)

Suggest three different combinations of two ADP methods in each case. Describe the advantages you expect from the combination of the respective two methods.